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CHANGES IN BASKETBALL SHOOTING PATTERNS AS A FUNCTION OF DISTANCE[1]

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Summary.—The effect of distance on the accuracy and movement form in basketball shooting was examined using a dynamic systems approach. Five male and five female young adults ($M = 26$ yr.) with no basketball experience beyond regular physical education shot a basketball 20 times at each of eight distances from 5 to 40 ft. (Natural condition). Also, they pretended to shoot the ball 5 times at each of the eight distances, for a total of 200 shots each (Pretend condition). In the Natural condition, shooting accuracy significantly decreased as shooting distance increased. Across both conditions and across four body components (feet position, hand position, trunk rotation, and jump height), the participants shifted from one movement pattern to another 86.3% of the time as shooting distance increased. The distances at which the transitions occurred were significantly shorter in the Natural than the Pretend condition for the feet and hand components but not the trunk and jump-height components. These results indicated that shooting a basketball at increasing distances can be portrayed as a dynamic system characterized by abrupt changes in at least four body components at critical distances.

In basketball, coaches, teachers, and researchers advocate certain movement patterns for shooting free throws, set shots, and jump shots. For example, for the free throw, players are encouraged either to have both feet even with the free-throw line (Amberry, 1996; Cooper, 1987) or to have the foot on the shooting-hand side (ipsilateral) to be slightly in front of the other, right up to the line (Cooper & Siedentop, 1969; Wissel, 1994). Also, players are admonished to keep the shooting hand above the head. In a biomechanical study, Hudson (1985) showed that female basketball players of varying skill levels released the ball on free throws about 23 to 30% of their standing height above their heads. Moreover, Brancazio (1981) reported that a higher release height minimizes the shooting angle and requires the least force, which also is the angle which maximizes the margin of error. As the distance from the basket is increased, players are recommended to drop the ball lower, flex the knees more, and rotate the trunk to increase the force on the ball (Cooper, 1987). Finally, on jump shots, players are told to adjust the height of the jump to the defense not to the actual distance of the shot (Wissel, 1994).

Most experts agree with these key points on shooting form, but exactly

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how does distance affect an individual's shooting pattern? A dynamic systems approach to studying movement development and performance (Kugler, Kelso, & Turvey, 1980; Thelen, 1995; Thelen & Smith, 1994; Thelen & Ulrich, 1991) indicates that as a key variable such as shooting distance is scaled up, a critical distance will be reached at which a preferred pattern will suddenly change and a new, more efficient pattern will emerge. Thelen and Smith (1994), in their recent book, explain seven steps of a dynamic systems approach to studying the development of cognition and action. For a nondevelopmental study, these seven steps can be adapted, rearranged, and reduced to five steps: (a) identify the dependent variables which best capture the collective behavior of the system, (b) identify potential "control parameters" or independent variables which may elicit changes in the "collective" or dependent variables, (c) manipulate the control parameters to generate phase transitions experimentally, (d) examine the preferred states of the collective variables at particular values of the control parameters, and (e) identify points of transition (Thelen & Smith, 1994, pp. 99-121).

In basketball, the dependent variables which best capture the collective behavior of the system are those that describe the shooting form (Step 1). Shooting accuracy also is an important dependent variable because it represents the outcome of the intended task, but accuracy is much less likely to reveal the dynamics—or the changes and stabilities—of the action system. Most previous studies on basketball shooting have focused exclusively on shooting accuracy, with just a few using movement form or mechanics as dependent variables (e.g., Hudson, 1985; Regimbal, Deller, & Plimpton, 1992; Satern, Messier, & Keller-McNulty, 1989).

Shooting distance is recognized by athletes and coaches as a prime factor that affects shooting accuracy and form and is perhaps the most important control parameter that varies within a game (Step 2), yet the dynamics of the effect of distance has not been published in the scientific literature. However, several studies have been reported examining the effect of basket height (Chase, Ewing, Lirgg, & George, 1994; Gabbard & Shea, 1980; Satern, *et al.*, 1989) and ball size (Chase, *et al.*, 1994; Regimbal, *et al.*, 1992; Satern, *et al.*, 1989) on shooting performance. The results of these studies consistently have shown that a basket lower than the regulation 10-ft. height significantly improves shooting accuracy, but that a ball smaller than a regulation men's ball (women's, junior standard, or intermediate size) does not have a significant effect on accuracy. These studies have focused on goal height and ball size, both of which can be easily modified by teacher or coach, but the effect of shooting distance which is primarily controlled by the player has not been assessed by researchers.

Thus, according to the dynamic systems perspective, as shooting distance is scaled up, the shooting form preferred at shorter distances should

"break down" or change at some critical distance (Steps 3, 4, and 5). This type of phase transition has been shown for many different skills, including bimanual tapping (Kelso, 1984), hand–foot tapping (Carson, Goodman, Kelso, & Elliott, 1995), handwriting (Burton & Dancisak, in press), throwing (Burton, Greer, & Wiese, 1992), and walking–running (Shapiro, Zernicke, Gregor, & Diestal, 1981). Research on rhythmic limb movements by Kelso and his colleagues (Carson, *et al.*, 1995; Kelso, 1984; Kelso, Scholz, & Schoner, 1986) has shown that a transition from one stable pattern to another is usually preceded by a short period of enhanced instability during which the system searches for a more optimal movement solution. Goldfield (1993), in his research on the onset of crawling in infants, has argued that variability or instability is associated with active exploration of more effective ways of solving a movement problem.

An individual is likely to use the same movement form from trial to trial within a particular performance condition when the best solution for the movement task is discovered through sufficient practice. Fitts and Posner (1967), in their classic three stages of motor learning, indicated that a consistent movement pattern or strategy is one of the indicators of a shift from the cognitive (beginning) to the associative (intermediate) stage. Thus, a consistent or stable pattern when the task and environmental constraints are perceived as being constant can be viewed as an important indicator that a person has moved beyond a novice or beginning skill level. This may occur almost immediately in simple tasks like bimanual tapping but require more time in more complex tasks like shooting a basketball for accuracy.

One of the problems in examining shooting form as a function of any control parameter is the lack of assessments of shooting form. Regimbal, *et al.* (1992) set four criteria for shooting skill technique (knees bent before shooting, eyes looking toward the rim, shooting elbow down, and hand follow through toward the rim), but the 1–3 scoring system for each criteria was not explained. A similar sequence of events is described by Amberry (1996), each of which could be evaluated with a "yes–no" system. These approaches focus on each body component separately but only offer one form option for each component. A possible model for creating a more complete assessment of shooting form is Roberton's component approach (1984) for evaluating changes in throwing patterns as children increase in age. In her system, there are three to four form levels for five body components, including the feet, trunk, backswing, humerus, and forearm.

The purpose of this study was to examine the effect of distance on basketball shooting accuracy and, more importantly, on movement patterns used to shoot the basketball. The primary dependent variables reflecting the collective behavior of the system were the movement pattern levels of the feet, hands, trunk, and jump height, using a system similar to Roberton's

component analysis of throwing (1984). The primary control parameter, or independent variable, chosen for this experiment was shooting distance, as it ranged from very close (5 ft.) to far away (40 ft.) from the basket, with the expectation that it would elicit changes in the shooting pattern levels and shooting accuracy. This prediction was based on the idea that a shooting form used to maximize accuracy at shorter distances will not be able to generate enough force to get the ball to the basket at longer distances. A second independent variable was whether the shooting pattern was performed with or without a ball. The "Pretend" or without-ball condition allowed testing the assumption that shooting pattern transitions are driven by increasing force demands as the shot distance increased.

The principal research question, whether increasing shooting distance from 5 to 40 ft. elicits changes in the shooting form and shooting accuracy of nonexpert players, was framed in terms of four hypotheses. First, individual participants should use the same shooting patterns on shots performed at the same distance away from the target in the Natural condition and the Pretend condition. Basketball players who have established a consistent pattern for the same shooting condition, e.g., Pretend at 15 ft., can be considered to have reached at least the "associative" or intermediate stage of motor learning for shooting (Fitts & Posner, 1967). Second, the movement patterns used by the different body components were expected to change abruptly as distance was scaled up to a critical value in the Natural and Pretend conditions. A maximum distance of 40 ft. was expected to be long enough to evoke pattern transitions in most individuals, based on data from a pilot study of high school basketball players of both sexes (Burton, Larson, & Liu, 1992). Third, the stability of the movement patterns used by individuals was predicted to be least at the distance just before the transition in both the Natural and Pretend conditions. Fourth, if the increasing force requirements at longer distances is an important factor in eliciting form changes, then the shooting pattern transitions should occur at shorter distances in the Natural or "with-ball" condition than in the Pretend condition due to the greater force needed with the ball.

Method

Participants

Five men and five women, ranging in age from 18 to 32 years, participated. All participants were right-handed and were university undergraduate or graduate students. None exhibited any perceptual or motor deficits that might interfere with the performance of the task, and none had any received special training in basketball shooting beyond regular physical education. The mean ages of the men and women were 25.2 and 26.6 yr., respectively.

This study was approved by the Institutional Review Board at the University of Minnesota, and informed consent was obtained from all participants.

Apparatus

A regulation basketball rim was mounted on a regulation rectangular wooden backboard at a regulation height of 10 ft. (3.05 m). White tape, 5.1-cm wide and 1.5-m long, marked eight locations extending out from the rim, perpendicular to the backboard. Each location was coded on the floor with a letter: A = 5 ft., B = 10 ft., up to H = 40 ft.

Two different basketball sizes were used, one for men and one for women. The men, ranging in hand widths (from thumb to little finger) from 19.5 to 23.0 cm, used an official men's basketball, while the women, ranging in hand widths from 19.0 to 20.0 cm, used an official women's basketball to minimize body-scale differences between the participants. The mean ratio of ball diameter to hand width was 1.16 for the men and 1.26 for the women, but, as mentioned in the introduction, previous studies have shown that ball size does not significantly affect shooting accuracy. The men's basketball weighed 595 g, had a circumference of 76.2 cm, and a diameter of 24.4 cm; the women's basketball weighed 539 g, had a circumference of 74.7 cm, and a diameter of 23.4 cm.

Two VHS videocameras mounted on tripods were used to record two right-side views of the participants shooting the basketballs. The cameras were placed perpendicular to the ball path, one directed toward shorter distances (Distances A–D) and the other toward longer distances (Distances E–H). The two cameras were both 6.4 m away from the shooting path, the first camera 15 ft. from the backboard, adjacent to Distance C, and the second 30 ft. from the backboard, adjacent to Distance F.

Procedure

When the participants first entered the gym, they read and signed an informed consent form. Next, they were encouraged to stretch out and warm up as much as they needed. The task procedures were fully explained to them, then they took 20 practice shots at distances between 10 and 15 ft.

All participants performed shots in two conditions, Pretend and Natural. In the Pretend condition, participants were instructed to perform the task using a preferred or "natural" foot position without a ball. In the Natural condition, participants were instructed to perform the task using a preferred or natural foot position with a ball. The participants began by performing one 5-trial block at each distance for the Pretend condition (40 shots), followed by 20 trials in blocks of five at each of eight distances for the Natural condition (160 shots). The Pretend trials were performed first rather than randomized with the Natural trials because the threat of shooting with a ball affecting shooting form without a ball was considered to be a

greater threat than fatigue or practice. The distance order of the blocks of five trials was randomized across the 10 participants as a control for possible systematic order or practice effects. Participants were encouraged to "make as many baskets as possible."

Design, Analysis, and Reliability

In this study, there was one within-subject independent variable (eight shooting distances) and one between-subject independent variable (sex). The primary dependent variable was comprised by levels of four components of a one-handed shooting pattern, including action related to the feet, hand, trunk, and jump height (see Table 1). A secondary dependent variable was the accuracy of the shots. Analyses of variance with repeated measures were used, followed by Fisher's Least Significant Difference (*LSD*) *post hoc* procedure when a main effect was found for a factor with more than two levels, applying an alpha level of .05 for each.

TABLE 1
LEVELS OF THE FOUR COMPONENTS OF THE ONE-HANDED SHOOTING PATTERN

Component	Level	Description
Feet	4	Contralateral foot forward, with distance between feet
	3	Contralateral foot forward, with overlapping feet
	2	Feet even
	1	Ipsilateral foot forward, with overlapping feet
	0	Ipsilateral foot forward, with distance between feet
Hand	3	Shooting hand over head
	2	Shooting hand at or above eyes up to the top of the head
	1	Shooting hand at or above chin up to eyes
	0	Shooting hand below chin
Trunk	1	Trunk rotation
	0	No trunk rotation
Jump Height	2	High jump (greater than 3 inches)
	1	Low jump (up to 3 inches)
	0	No jump at all (at least one foot on ground)

The principal investigator and one trained undergraduate kinesiology student each independently coded the shooting patterns of the 10 participants on all trials. The shooting patterns were coded according to levels of four body components: feet (five levels), hand (four levels), trunk (two levels), and jump height (three levels) (see Table 1). The highest level within each component did not indicate "proper" or "best" form; the levels represented the categories of movement that are possible within each component. The five levels of the feet specify the position of one foot relative to the other; the four levels of the hand specify the position of the shooting hand relative to other body parts; the two levels of the trunk indicate whether the

trunk rotated or not; and the three levels of the jump height specify whether both feet left the floor, and if so, whether the jump exceeded 3 inches or not. The 3-in. criterion was used to objectively discriminate between a "very low" jump just off the ground from a higher jump.

Interrater reliability of the coding of the four components was computed across all trials by all participants ($N = 200$ trials × 10 participants = 2,000). The proportion of agreement between the two raters for all coded patterns were .97, .95, .99, and .97 for feet, hand, trunk, and jump-height components. For all subsequent analyses, the data coded by the principal investigator only was used.

Results

Shooting Accuracy

Shooting accuracy decreased as distance increased for the Natural condition, as shown in Fig. 1. Note that there were no data for shooting accu-

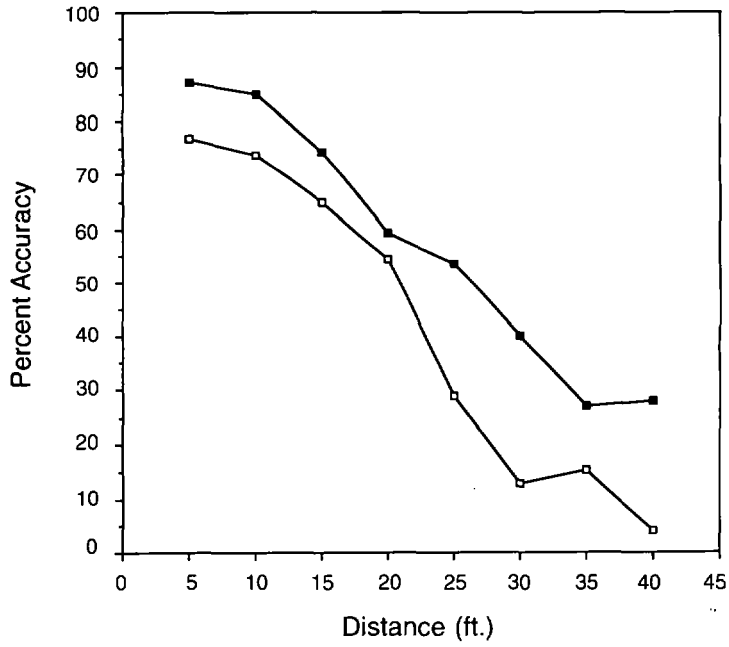

FIG. 1. Percent shooting accuracy in the Natural condition by sex and distance: (■) Men (□) Women

racy for the Pretend condition because the ball was not actually shot. A two-way analysis of variance (Distance × Sex) with repeated measures indicated that the men (57%) were significantly more accurate than the women

(42%; $F_{1,8}=5.80$, $p<.05$) and that accuracy significantly decreased with increasing distance ($F_{7,56}=80.45$, $p<.001$).

Stability of Shooting Patterns at Particular Distances

The first hypothesis was concerned with the stability of the throwing patterns used by each individual over the 20 trials at each of eight distances in both conditions. The participants showed remarkable stability in all body components across distances and conditions. The mean percentage of the most common component levels across all distances and the two conditions for individual participants was between 90.5% to 97.6% for the four components, indicating that the component levels were stable (see Table 2). The mean percentages were higher for men than women and higher in the Pretend than the Natural condition for all four components.

TABLE 2
MEAN PERCENTAGE AND STANDARD DEVIATIONS OF THE MOST STABLE PATTERN FOR EACH COMPONENT BY CONDITION AND SEX

Condition/Sex	Feet		Hand		Trunk		Jump	
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Pretend	97.2	8.4	98.2	6.5	100.0	0.0	95.4	10.6
Men	96.4	9.2	97.4	8.1	100.0	0.0	96.5	10.0
Women	98.0	7.6	99.0	4.4	100.0	0.0	94.4	11.3
Natural	85.9	18.6	91.6	14.8	95.3	10.8	85.4	17.6
Men	88.8	18.2	97.4	7.5	97.8	7.2	89.6	17.3
Women	83.0	18.7	85.9	17.9	92.8	13.1	91.3	17.2

A three-way analysis of variance (Sex × Condition × Distance) with repeated measures showed that the effect of sex on the mean percentage of the most common pattern was statistically significant only for the hand component ($F_{1,8}=11.6$, $p<.01$). The effect of condition was significant for all four components: feet ($F_{1,8}=15.7$, $p<.01$), hand ($F_{1,8}=24.2$, $p<.01$), trunk ($F_{1,8}=11.9$, $p<.01$), and jump height ($F_{1,8}=10.0$, $p<.05$). Also, there was a significant interaction for sex × condition for the hand component ($F_{1,8}=24.2$, $p<.01$). This interaction was accounted for by the mean stability being the same for men in the two conditions, but much higher for women in the Pretend than the Natural condition (see Table 2). Neither the effect of distance nor any of the related interactions were statistically significant for any component.

Eight of the 10 individual participants demonstrated the same component level on at least 50% of the trials for all components at all distances in both conditions. The most common pattern fell below 50% for one woman in the Natural condition for the feet component at 15 ft. and for the jump-height component at 20 ft., and for one man in the Natural condition for the feet component at 30 ft.

Transitions from One Stable Component Level to Another

A transition refers to an abrupt qualitative change or reorganization of a movement pattern. For instance, the forward foot may change from ipsilateral to contralateral or the shooting hand may change from high position (above nose) to low position (below nose) as distance is scaled up. Thus, in the second hypothesis, significant qualitative changes of the movement patterns were predicted from one stable level to another across the distances in both Natural and Pretend conditions. Across both conditions, both sexes, and four body components, a shift from one pattern to another without regression back to a previous pattern occurred 86.3% of the time, indicating significant qualitative changes of the movement patterns from one stable level to another across the distances. The transition percentage was lowest for women (75.0%), the Pretend condition (80.0%), and the jump-height component (80.0%), but chi-squared analyses on the actual frequencies showed that the differences across sex, condition, and component were not statistically significant.

As distance increased, the general sequences were from (a) ipsilateral to contralateral forward foot position, (b) high to low hand position, (c) no trunk rotation to trunk rotation, and (d) low to high jump. The individual transition sequences observed for each component by condition are detailed in Table 3. For the feet component, the most common transition sequences

TABLE 3
SEQUENCE OF PATTERNS USED FOR EACH COMPONENT BY INDIVIDUAL PARTICIPANTS (NUMBER INDICATED IN PARENTHESES) BY CONDITION

Component	Pretend	Natural
Feet	2-4 (4)	2-4 (6)
	1-4 (4)	1-4 (4)
	2 only (2)	
Hand	1-0 (2)	1-0 (2)
	2-0 (1)	2-1 (1)
	2-1-0 (2)	2-1-0 (1)
	3-2-0 (2)	3-0 (2)
	3-2-1 (1)	3-2 (1)
	2-3-2-0 (1)[a]	3-2-0 (2)
	1-2-3 (1)[b]	3-2-0-1 (1)[a]
Trunk	0-1 (8)	0-1 (10)
	0 only (2)	
Jump	0-1 (2)	0-1-2 (4)
	0-1-0 (1)[a]	0-2 (3)
	0-2 (2)	1-2 (1)
	1-2 (4)	0-1-2-0 (1)[a]
	2-0 (1)	0-1-2-1 (1)[a]

[a] Sequence with reversal of pattern direction. [b] Sequence with transitions going in the direction opposite the standard direction.

were from even to contralateral foot (2–4) and from ipsilateral to contralateral (1–4), shown by 90.0% of the participants across both conditions. There were many variations in the hand components, but 85% of the transition sequences went from higher to lower hand position.

The most common hand sequences used to shoot the basketball were from above the head to below the chin (3-2-0, 3-1-0, and 3-0) shown at four of the 10 distances across both conditions, and from the forehead to below the chin (2-1-0, 2-0) shown at two of the 10 distances. A change back to previous hand patterns occurred once (10%) in both conditions, and a sequence from a lower to a higher position also was shown once in the Pretend condition.

All participants demonstrated a change in trunk pattern from no rotation to rotation in the Natural condition, and only two women showed no trunk rotation at any distance in the Pretend condition. Transitions from low to high jump were found 80% of the time. The most common jump-height sequence across both conditions was from no jump at all to high jump (0-1-2 and 0-2) (45%). A change back to previous jump-height components was observed once in the Pretend condition (10%) and twice in the Natural condition (20.0%).

Instability and Critical Transition Distance

A significant drop in pattern stability just prior to the critical distance at which a transition occurred was predicted by the dynamic systems approach in the third hypothesis. The distance of least stability of the movement patterns for individual participants were determined and categorized into four possibilities: just before the critical distance at which a shift from one pattern to another occurred without regression back to a previous pattern, right at the critical distance, an abrupt transition without instability, and no transition. Fig. 2 shows that the point of least stability in the Natural condition occurred at the distance right before the transition 80.0% of the time; however, in the Pretend condition, an abrupt transition without any instability was observed 70.0% of the time.

Effect of Condition on Transition Distance

According to last hypothesis, individuals were expected to show pattern transitions at shorter distances in the Natural condition than in the Pretend condition. Note that if a person did not show a transition, the distance was listed as 45 ft., the next possible increment at which a transition could have occurred. The data showed that the mean transition distance averaged across body components was 23.5 ft. for the Natural condition and 28.4 ft. for the Pretend condition. The mean transition distance was 30.5, 24.5, 26.8, and 22.0 for the feet, hand, trunk, and jump-height components, respectively. Four separate two-way analyses of variance (condition × sex) with repeated

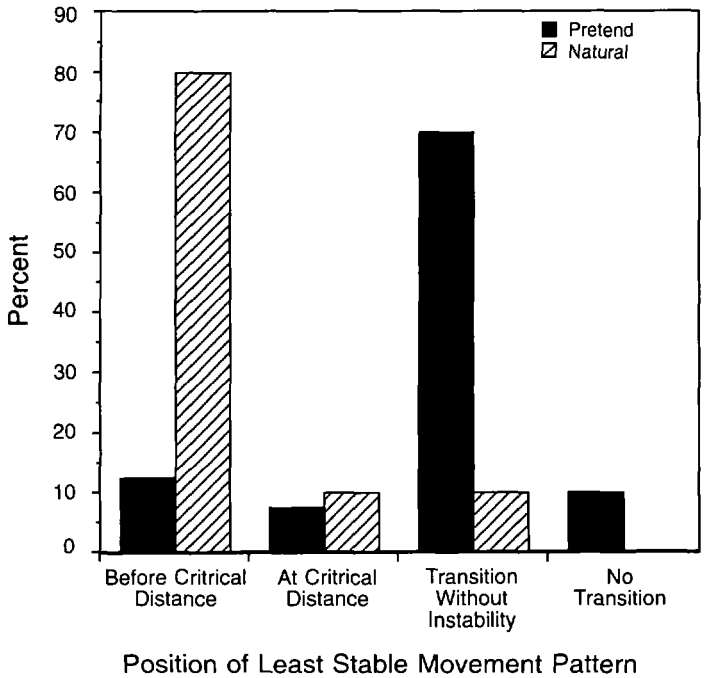

FIG. 2. Relative frequency of the position of the least stable movement patterns for the Pretend and Natural conditions

measures showed significantly shorter transition distances for the Natural than the Pretend condition for the feet and hand components ($F_{1,8}=11.20$ and 6.88, $ps<.05$) but not the trunk and jump-height components. For the feet component, women had significantly longer transition distances than men ($F_{1,8}=10.13$, $p<.05$). For the hand component, a significant interaction of condition × sex ($F_{1,8}=6.88$, $p<.05$) was accounted for by women showing shorter transition distances in the Natural condition and the men showing the same mean transition distance in both conditions.

DISCUSSION

Stability of Shooting Patterns at Particular Distances

As predicted, the movement patterns used to shoot at particular distances were quite stable, with the mean percentage of the most common component levels across all distances and both conditions ranging between 90.5% to 97.6% for the four components. Even with the perturbation of randomly changing shooting distance every five shots, the participants were judged as having the same component pattern 90% of the time when shoot-

ing at the same distance. The data for individual participants showed that all of them used consistent movement patterns at particular distances, indicating that they had reached what Fitts and Posner (1967) called the associative (or intermediate) stage of skill learning.

The mean stability was significantly lower in the Natural condition than in the Pretend condition for all four body components. The higher stability in the Pretend condition compared to the Natural condition may be best explained by the lack of outcome feedback in the Pretend condition. Because no ball was actually shot, the performers missed no baskets, and they had no need to adjust their shooting patterns to improve performance.

An analysis of the component level combinations in the Natural condition showed that, when a contralateral step was used (Level 3 or 4, $n=673$), trunk rotation occurred 99.1% of the time, but when an ipsilateral step was used (Level 0 or 1, $n=321$), trunk rotation occurred only 62.9% of the time.

The greatest instability was found for women with the hand component in the Natural condition, as indicated by a significant interaction of sex × condition. The significant difference for women between the Natural (85.9%) and the Pretend (99.0%) conditions indicated that the mass of the ball was a key factor, and lack of a condition effect for men suggested that upper-body strength limitations in the women may have caused them to continue to try different movement patterns. Given enough practice, this interaction of sex × condition would likely disappear.

Transitions from One Stable Component Level to Another

The second hypothesis was supported by the results which showed that the 10 participants switched from one stable component level to another 86.3% of the time across the two conditions and the four body components. As distance increased, the new component patterns allowed the performers to generate more force in their shots. The shift from an ipsilateral to a contralateral step freed the body to add trunk rotation and other motions to the shot. The dropping of the shooting hand from a high to low position allowed the hand to accelerate the ball over a longer distance, resulting in greater force. Similarly, the addition of trunk rotation and a higher jump to the shooting motion allowed force to be summed across the actions of multiple joints. The percentage of transitions was lower in the Pretend condition (80.0%) than in the Natural condition (92.5%). Although not significantly lower, it again points to performance outcomes driving changes in movement form.

Only six of the 76 (7.9%) transition sequences did not follow the sequences described above. In two hand sequences and three jump-height sequences, the individuals reverted to a previously used pattern as distance continued to increase. And, in one hand sequence, the hand moved from low

to high as distance increased. These few exceptions indicated that, across both conditions, almost all of the individuals arrived at the same general solutions for the changing force and accuracy demands of the shooting task.

Although the transition sequences within each body component were very similar across participants, the four-part, component-level combinations appeared to be very idiosyncratic. Of the 120 possible component-level combinations (5 feet levels × 4 hand levels × 2 trunk levels × 3 jump-height levels), 68 were shown in the Natural condition across all participants. One combination (4-0-1-2) accounted for more than 10.3% of the 1,600 trials, and the 38 combinations each occurring less than 1% of the time also accounted for 10.3% of the total. The remaining 29 combinations, each occurring from 1.0 to 6.6% of the time, accounted for 79.4% of the total. The single most common pattern at a particular distance ranged from a high of 35.5% (4-0-1-2 at 40 ft.) to a low of 12.0% (2-0-0-0 at 20 ft.). Thus, few participants used the same total body solutions, even at the same distances. The wide variety of component combinations observed in just these 10 participants validated the use of a component approach to describe the shooting motion. An assessment system offering just four to even 10 total-body patterns would miss many of the shooting differences between individuals and within individuals across distances.

Instability and Critical Transition Distance

The prediction that pattern stability would be lowest at the distance just prior to the critical transition was supported by the data. This result, first demonstrated in the studies on rhythmic limb movements by Kelso and colleagues (Carson, *et al.*, 1995; Kelso, 1984; Kelso, *et al.*, 1986), indicates that the transition to a new pattern is preceded by a period of exploring new solutions to the movement problem, switching from one pattern to another, even while performing at the same distance. An abrupt transition from one pattern to another without instability was shown only 10% of the time for the Natural condition, but 70% of the time for the Pretend condition. This large difference suggests again that feedback about performance plays a very important role in the dynamics of movement. Specifically, the urge to change to a new pattern at a particular distance may be validated by maintained or improved shooting accuracy, but even after just one miss an individual may revert to an old pattern and vacillate between two or more patterns, depending on the performance. As simulated in the Pretend condition, many coaches and instructors in basketball (and other sports, too) teach their players and students to trust their shot, to ignore the outcome of the shot, and not to make adjustments after a miss. For example, Amberry (1996), who holds the world's record for consecutive free throws, offered this advice: "Never say to yourself, 'You almost missed it short. Shoot this one

longer!' Don't compare your next shot to your previous one. That shot is history. It's gone" (pp. 68-69).

Effect of Condition on Transition Distance

The fourth hypothesis was based on the concept that, if shooting pattern transitions are driven by the demands for increasing force as the shot distance increases, then the transitions should occur at shorter distances in the Natural condition than in the Pretend condition. The results indicate that the mean transition distances were significantly longer in the Pretend condition than in the Natural condition for the feet and hand components. However, this delayed transition in the Pretend condition also could have been related to the lack of performance feedback in this condition. If the performers never missed a shot, there was no need to switch to a new pattern. To control for the confounding effects of using the ball and receiving performance feedback, another experimental condition is needed in which the ball is shot using a natural pattern, but the performers would not receive any feedback regarding the outcome of the shot.

Conclusion

The results of this study indicate that shooting a basketball at increasing distances can be portrayed as a dynamic system. As the difficulty of the task increases, as shown by significantly decreasing shooting accuracy, the form of the shooting motion changes for individuals at critical distances. The most common form transitions were from ipsilateral to contralateral forward foot position, from high to low hand position, from no trunk rotation to trunk rotation, and from low to high jump. At the same distances, shooting patterns were quite stable. In the Natural condition, increasing instability of form for shots at the same distance indicated that a transition to a new form was imminent. In the Pretend condition, little instability was shown, even right before a transition, which points to the importance of feedback about performance in adjusting movement form. In other words, as a player begins to see a drop in shooting accuracy, new shooting pattern alternatives may begin to be used.

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